

Math 421

Friday, October 10

Announcements

- Homework 5 due **today**
- Exam 1 – 10/15 – 5:45-7:15pm
 - If you have a conflict, email me (if you haven't already) and **fill out makeup exam request form** (linked from Exam Information page on Canvas) ASAP!
 - If you are a McBurney student, sign up to take the exam at the testing center through RegisterBlast.

Know: $\lim \frac{1}{n} = 0$ $\lim \frac{1}{n} \cdot \lim \frac{1}{n} = \lim \left(\frac{1}{n^2} \right) = 0 \cdot 0 = 0$

9.7 Theorem (Basic Examples)

a) $\lim \left(\frac{1}{n^p} \right) = 0$ for $p > 0$.

$\lim \frac{1}{n^{1/4}} = \lim \left(\frac{1}{n} \right)^{1/4} \rightsquigarrow$

can't use our algebraic
limit laws.

b) $\lim a^n = 0$ if $|a| < 1$.

c) $\lim (n^{1/n}) = 1$.

d) $\lim (a^{1/n}) = 1$ for $a > 0$.

Activity – Prove 9.7 a)

Theorem. $\lim \left(\frac{1}{n^p} \right) = 0$ for $p > 0$.

Discussion.

$$\left| \frac{1}{n^p} - 0 \right| < \varepsilon \iff \frac{1}{n^p} < \varepsilon \iff n > \left(\frac{1}{\varepsilon} \right)^{1/p} \quad \text{Set } N = \left(\frac{1}{\varepsilon} \right)^{1/p}$$

Proof. Let $\varepsilon > 0$. Set $N = \left(\frac{1}{\varepsilon} \right)^{1/p}$. Assume $n > N$, then

$$\left| \frac{1}{n^p} - 0 \right| < \left| \frac{1}{\left[\left(\frac{1}{\varepsilon} \right)^{1/p} \right]^p} \right| = \varepsilon.$$

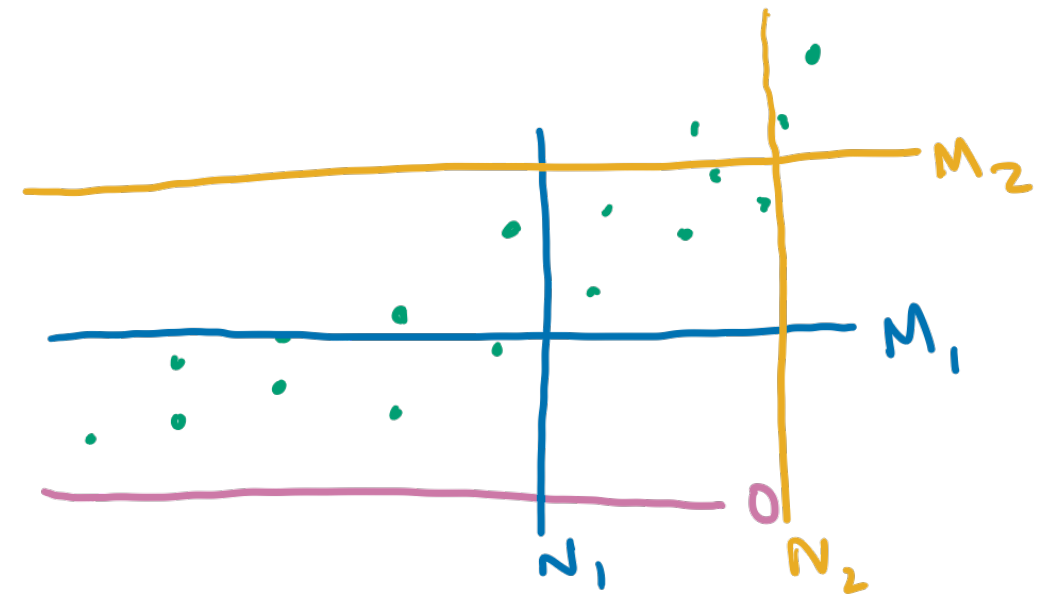
Since $\varepsilon > 0$ was arbitrary, $\lim \left(\frac{1}{n^p} \right) = 0$ for $p > 0$.

Diverging to $\pm\infty$

Definition 9.8

- $\lim s_n = +\infty$ provided for each $M > 0$ there is a number N such that $n > N$ implies $s_n > M$.
- $\lim s_n = -\infty$ provided for each $M < 0$ there is a number N such that $n > N$ implies $s_n < M$.

playing the
role of ϵ



Terminology: A sequence (s_n) has a **limit** or the **limit exists** if (s_n) converges to a real number or diverges to $\pm\infty$.

$((-1)^n)$ diverges and does not have a limit.

Example/Activity

Example: $\lim n^2 = +\infty$

$$n^2 > M \iff n > \sqrt{M} \quad \text{set } N = \sqrt{M}$$

Let $M > 0$. Set $N = \sqrt{M}$. Assume $n > N$, then

$$n^2 > (\sqrt{M})^2 = M.$$

Since $M > 0$ was arbitrary, $\lim n^2 = +\infty$.

Activity: $\lim(\sqrt{n} + 7) = +\infty$

$$\text{set } N = (M - 7)^2.$$

$\lim s_n = +\infty$ if for all $M > 0$ there exists N
 such that
 for all $n > N$,
 $s_n > M$.
 (1, 2, 0, 3, 4, 0, 5, 6, 0, ...)

Activity

1. What does it mean for an unbounded sequence to not diverge to $+\infty$ or $-\infty$, i.e. what is the negation of the definition?

There exists $M > 0$ such that for all N , there
 exists $n > N$ such that $s_n \leq M$.

Sketch

2. Prove that the unbounded sequence $s_n = (-1)^n n$ does not have $+\infty$ as a limit. What proof technique should you use?

Assume N $M > 0$, there exists contradiction
 $s_n > M$. if n is odd $s_n = -n > M$

3. Modify your proof to show that $s_n = (-1)^n n$ does not have $-\infty$ as a limit.

Caution!

The limit laws we've previously stated do not apply to sequences that have infinite limits, we need to prove these separately.